

To:

The Editor-in-Chief and Editorial Board

Robotica

Cambridge University Press

Subject: Submission of Original Research Article: “*Danger-State Detection in Human-Robot Collaboration Using Deep Learning*”

Dear Editor-in-Chief and Editorial Board Members,

We are pleased to submit our original research manuscript entitled “**Danger-State Detection in Human-Robot Collaboration Using Deep Learning**” by Ameur Gargouri, Mohamed Karray, and Mohamed Ksantini for consideration for publication as a regular research article in *Robotica*.

Context and Motivation

As industrial automation transitions toward Industry 5.0, physical Human-Robot Collaboration (pHRC) has become essential for flexible and intelligent manufacturing. Safe co-existence and direct physical interaction between human workers and collaborative robots (cobots) demand continuous, real-time assessment of workspace hazards in compliance with standards such as ISO/TS 15066. However, current safety mechanisms face a severe **false-alarm trade-off**: overly sensitive detectors frequently command unwarranted emergency stops that disrupt production cadence and compromise worker trust, whereas insufficiently sensitive systems risk severe operator injury from high-impact collisions, clamping, or crushing events.

Conventional collision avoidance systems typically rely on costly multi-axis force/torque transducers, external multi-camera vision rigs that are prone to line-of-sight occlusions and lighting variations, or disturbance observers that require mathematically exacting nominal dynamic models. Utilizing wearable inertial measurement units (IMUs) provides an attractive, cost-effective, and occlusion-free alternative. Nevertheless, current literature lacks a systematic empirical comparison of modern deep learning sequence architectures on wearable sensor streams, as well as deterministic mechanisms designed specifically to curtail false alarms.

Key Contributions of the Manuscript

This work bridges this gap by proposing a complete, end-to-end deep learning framework tailored for robust, high-accuracy, and low-false-alarm danger-state detection:

1. **GPU-Accelerated Kinematic Expansion:** We develop an on-the-fly kinematic derivative pipeline running entirely on GPU, deriving joint velocities and accelerations from raw 65-channel IMU position streams to generate 195 feature channels without additional hardware sensors or offline storage overhead.
2. **Systematic Benchmark of Modern 1D Architectures:** We perform a rigorous comparative evaluation of five distinct 1D architectures—ConvNeXt 1D, Temporal Convolutional Networks (TCN), MLP-Mixer 1D, Transformer 1D, and InceptionTime—with and without Test-Time Augmentation (TTA). ConvNeXt 1D and TCN achieve state-of-the-art macro-F1 scores of 98.68% and 98.62%, respectively, with danger precision exceeding 98% and recall surpassing 97%.
3. **Dedicated False-Alarm Reduction Post-Processing:** We design a tailored temporal debouncing stage integrating a 3-tap median filter with decision threshold optimization. This mechanism reduces false alarms by **87.5%** while maintaining danger detection recall above 97.1%, establishing a viable operating point for industrial deployment.
4. **Rigorous Benchmark Validation:** The framework is validated via 5-fold stratified cross-validation over more than 69,000 temporal windows gathered from 60 human subjects on the DASIG benchmark, verifying statistical robustness across diverse operating profiles.

Relevance to the Scope of *Robotica*

Robotica is a premier international forum dedicated to the multidisciplinary science and practice of robotics, sensor systems, and artificial intelligence in automation. Our manuscript aligns closely with the core interests of *Robotica*'s readership:

- **Human-Robot Interaction & Safe Cobots:** Formulates an actionable solution for certified human safety and collision mitigation in shared industrial workspaces.
- **Applied AI & Machine Learning in Robotics:** Demonstrates how modern 1D deep learning backbones combined with domain-specific signal engineering solve time-critical robotics perception tasks.
- **Sensor Perception & Wearable Systems:** Explores wearable inertial telemetry as an economical, occlusion-free alternative to traditional external vision and force sensors in dynamic industrial environments.

Author Declarations

- This manuscript is an original contribution, has not been published previously, and is not currently under consideration for publication elsewhere.
- All authors have read and approved the final manuscript and agree with its submission to *Robotica*.
- The authors declare no conflicts of interest or competing financial interests.
- The data analyzed during this study are available from the corresponding author upon reasonable request.

Thank you very much for your time, consideration, and management of the peer-review process. We look forward to the reviewers' constructive feedback.

Sincerely,

Ameur Gargouri

Ph.D. Student / Researcher

ATISP Lab, ENET'Com, University of Sfax & ESME Research Lab

Email: `ameur.gargouri.doc@enetcom.usf.tn`

On behalf of all authors:

Dr. Mohamed Karray (ESME Research Lab, Corresponding Author)

Prof. Mohamed Ksantini (ATISP Lab, ENET'Com, University of Sfax)